RoamScope | Windows Setup & Roaming Test

Windows 10/11 • PowerShell • Teammate testing guide

Scope: These steps use the existing RoamScope Python package. Windows collector behavior has not yet been verified on the target laptop. Complete the preflight checks before the live walk. No macOS Swift helper is required.

1. Install and verify Python

Install an IT-approved Python 3 release from https://www.python.org/downloads/windows/ . Enable Add Python to PATH if offered. Open a new PowerShell window:

```
py --version
```

If the Python launcher is unavailable, try **python --version**. Use the working command consistently in the following steps.

2. Extract the RoamScope package

Extract the supplied ZIP into a new folder (do not mix with previous builds). Open PowerShell in the folder containing **roamscope.py**, **requirements.txt** and the **roamscope** directory.

```
dir roamscope.py, requirements.txt
```

3. Create environment and install dependencies

```
py -m venv .venv
.\.venv\Scripts\python.exe -m pip install -r requirements.txt
```

Using the environment interpreter directly avoids PowerShell activation and execution-policy changes.

4. Confirm Windows Wi-Fi telemetry

Connect to the intended corporate SSID, then run:

```
netsh wlan show interfaces
.\.venv\Scripts\python.exe roamscope.py --doctor
```

Verify the backend is **windows** and a real connected **BSSID** is available. Record the SSID, BSSID, channel and RSSI/signal information shown. Some fields depend on the adapter and driver. **Do not start the real survey if the BSSID is missing.**

5. Run a live roaming survey

From the RoamScope root directory, run:

```
.\.venv\Scripts\python.exe roamscope.py --duration 0 --interval 2 --csv
```

Walk the planned route through multiple AP coverage areas. At the end, press **Ctrl+C** to request a clean stop and report generation. The duration-zero and Ctrl+C workflow must be verified on this Windows build; if it does not finalize, repeat with a fixed duration such as **--duration 300**.

6. Inspect and share results

Check the console for the generated output path, typically under **roamscope-out**. Review the HTML report and export the roaming and scan CSV files. If available, capture the exact survey start/end times and the device Wi-Fi MAC address.

7. Cross-check against Mist

Filter Mist client roaming events to the same device and survey time window. Compare the event count, timestamps, source and destination BSSIDs, and channels/bands. Matching counts alone do not establish event-by-event agreement. Sampling every two seconds can miss brief transitions.

Troubleshooting

py not found: Try python or reinstall an approved Python with the launcher enabled. **Missing packages:** Repeat pip installation using the .venv interpreter. **Wrong folder:** Run dir and locate roamscope.py. **No BSSID:** Confirm the machine is associated to Wi-Fi, inspect netsh output, and share the doctor output before continuing. **No report:** Preserve console output and try a finite duration.

Safety: RoamScope is a passive client telemetry test; it does not require changing AP configuration. Follow company software installation and data-sharing policies.